Path Energy Traffic Ratio API (PETRA)

draft-petra-green-api-01

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Motivation

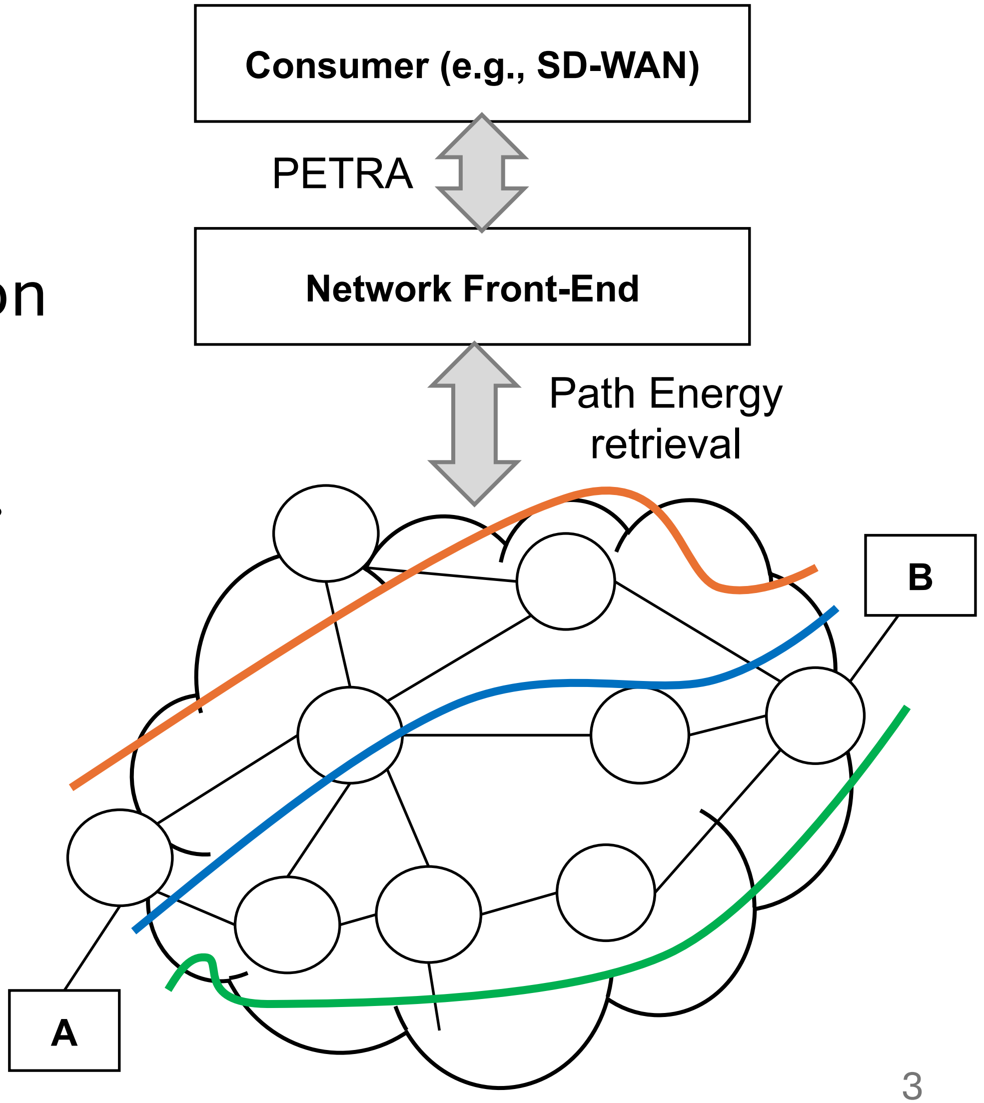
- Provide visibility about energy consumption in a path
  - Metrics such as power consumption between source and destination (potentially related with throughput)
- Define an API that can provide such information
  - Using well-known architectures and schemas (e.g. YANG)
- This information can be consumed externally (e.g., SD-WAN customers) or internally (e.g., for operator optimization purposes)

- Version -00 of the draft presented at GREEN WG in IETF 121
  - Earlier versions presented in PANRG

Path Energy Traffic Ratio API (PETRA)

- There can be multiple paths between origin and destination
- Energy consumption might be different on each path: dependent on device characteristics and architecture, transceiver bit rate, number of hops, etc.

- API Query:
  - <src-IP, dst-IP, throughput, ...>
- API Response:
  - <watts-per-gigabit, carbon-intensity, ...>



Changes between -00 and -01

- Explicitly called out the API is agnostic to the underlay infra
- Linked to Service Models in SDN context as described in RFC8309
- Expanded on recursive usage, particularly double counting
- Possible placement of PETRA in the Green Framework

Green Framework
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PETRA

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+--------------------------------------------------------------------+
|                                                                    |
|                   (3) Network Domain Level                         |
|                                                                    |
+--------------------------------------------------------------------+

(a)              (b)              (c)
Inventory        Monitor       +- DataSheets/DataBase
Of identity      Energy        |  and/or via API
and Capability   Efficiency    |  Metadata and other
      ^                ^       | device/component/network
      |                |       | related information:          (g)
      |                |       |                               PETRA
      |                |       |  .Power/Energy related metrics  API
      |                |       |  .information                    ^
      |                |       |  .origin of Energy Mix           |
      |                |       |  .carbon aware based on location |
      |                |       |                                  |
      |                |       v                                  |
+--------------------------------------------------------------------+ |
|                                                                  | |
|       (2) controller (collection, compute and aggregate?)        +-+
|                                                                  |
+--------------------------------------------------------------------+
                ^                       ^                       ^ |
      (d)       |       (e)             |    (f)                | |
      Inventory |       Monitor power   |    Control            | |
      Capability|       Proportion      |    (Energy saving     | |
                |       Energy efficiency|   Functionality      | |
                |       ratio, power    |    Localized mgmt/    | |
                |       consumption,    |    network wide mgmt) | |
                |       etc)            |                       | |
                |                       |                       | v
+--------------------------------------------------------------------+
|                                                                    |
|                      (1) Device/Component                          |
|                                                                    |
| +---------+  +-----------+  +-----------------+  +---------------+ |
| | (I)     |  | (II)      |  | (III)           |  | (IV)          | |
| |         |  |           |  | Legacy          |  | 'Attached'(PoE| |
| | Device  |  | Component |  | Device          |  | end Point)    | |
| |         |  |           |  |                 |  |               | |
| +---------+  +-----------+  +-----------------+  +---------------+ |
+--------------------------------------------------------------------+
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Next steps

- Gather and incorporate WG feedback
- Explore incentive scenarios